

Module 04: Saving Biodiversity

Questions & Answers

1. Define biodiversity. Explain the three levels of biodiversity with suitable examples. Discuss the importance of biodiversity.

Definition of Biodiversity

Biodiversity refers to the variety and variability of living organisms present on Earth. It includes different plants, animals, microorganisms, their genes, and the ecosystems in which they live. Biodiversity is essential for maintaining ecological balance and supporting life.

Three Levels of Biodiversity

1. Genetic Diversity

It refers to the variation in genes among individuals of the same species.

Examples:

- Different varieties of rice, wheat, and mangoes
- Different breeds of dogs and cattle
- Genetic differences among humans

Importance:

- Helps species adapt to environmental changes
- Increases resistance to diseases
- Supports evolution and survival

2. Species Diversity

It refers to the variety of different species present in a particular region or ecosystem.

Examples:

- Forests containing trees, birds, insects, and mammals
- Coral reefs having many fish and marine species

Importance:

- Maintains food chains and food webs
 - Supports ecosystem stability
 - Prevents ecological imbalance
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3. Ecosystem Diversity

It refers to the variety of ecosystems or habitats found in a geographical area.

Examples:

- Forests
- Grasslands
- Deserts
- Wetlands
- Oceans and rivers

Importance:

- Provides different habitats for organisms
- Supports various life forms
- Maintains nutrient and water cycles

Importance of Biodiversity

1. Ecological Importance

- Maintains ecological balance
- Supports pollination and decomposition
- Regulates climate and water cycle

2. Economic Importance

- Provides food, timber, fuel, and fiber
- Source of medicines and industrial products
- Supports agriculture and fisheries

3. Social and Cultural Importance

- Supports livelihoods of local communities
- Has religious and cultural value

4. Scientific Importance

- Useful in research, biotechnology, and genetic improvement

2. What is Biodiversity Under Threat? Explain the major threats to biodiversity such as habitat loss, climate change, pollution, overexploitation, and invasive species.

Introduction

Biodiversity under threat means the gradual decline in the variety of plants, animals, microorganisms, and ecosystems due to natural and human-induced activities. Rapid loss of biodiversity affects ecological balance, food security, and environmental sustainability.

Major Threats to Biodiversity

1. Habitat Loss

Habitat loss is the destruction or fragmentation of natural habitats where organisms live.

Causes:

- Deforestation
- Urbanization
- Agricultural expansion
- Industrial development
- Construction of roads and dams

Effects:

- Loss of shelter and food for wildlife
- Migration of species
- Extinction of rare species

Example:

Clearing forests for agriculture reduces wildlife populations.

2. Climate Change

Climate change refers to long-term changes in temperature, rainfall, and weather patterns.

Effects on Biodiversity:

- Species shift to cooler areas
- Changes in breeding and migration patterns
- Coral bleaching in oceans
- Extinction of climate-sensitive species

Example:

Melting polar ice affects polar bears and marine life.

3. Pollution

Pollution of air, water, and soil harms living organisms and ecosystems.

Examples:

- Industrial waste in rivers
- Oil spills in oceans
- Plastic pollution affecting marine animals
- Pesticides harming birds and insects

Effects:

- Toxicity to organisms
- Reduced reproduction and growth
- Death of aquatic and terrestrial species

4. Overexploitation

Overexploitation means excessive use of natural resources beyond sustainable limits.

Examples:

- Overfishing
- Hunting and poaching of animals
- Excessive cutting of trees
- Overharvesting medicinal plants

Effects:

- Population decline of species
 - Ecological imbalance
 - Resource depletion
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5. Invasive Species

Invasive species are non-native species introduced into an ecosystem that harm native species.

Effects:

- Compete for food and space
- Spread diseases
- Disturb local food chains
- Reduce native biodiversity

Example:

Water hyacinth blocks water bodies and affects aquatic life.

3.Explain the effects of biodiversity loss on environment, economy, and human life.

Introduction

Biodiversity includes the variety of plants, animals, microorganisms, and ecosystems on Earth. Loss of biodiversity occurs when species become extinct, habitats are destroyed, and ecosystem balance is disturbed. This creates serious impacts on the environment, economy, and human life.

1. Effects on Environment

a) Ecological Imbalance

Loss of one species can disturb the food chain and food web, affecting many other organisms.

b) Loss of Ecosystem Stability

Healthy ecosystems become weak and less able to recover from natural disasters or climate change.

c) Soil Degradation

Reduction of vegetation leads to soil erosion, loss of fertility, and desertification.

d) Water Cycle Disturbance

Forests and wetlands regulate rainfall and groundwater recharge. Their loss causes floods and droughts.

e) Climate Change

Forests absorb carbon dioxide. Loss of forests increases greenhouse gases and global warming.

2. Effects on Economy

a) Loss of Natural Resources

Biodiversity provides timber, fish, fuel, fibers, and raw materials. Loss reduces resource availability.

b) Decline in Agriculture

Loss of pollinators like bees reduces crop production. Loss of genetic diversity affects crop improvement.

c) Impact on Fisheries

Decline in marine biodiversity reduces fish stocks and affects fishing industries.

d) Loss of Tourism Income

Wildlife tourism and ecotourism decline when biodiversity decreases.

e) Increased Disaster Costs

Environmental degradation leads to floods, droughts, and storms causing economic losses.

3. Effects on Human Life

a) Food Insecurity

Loss of biodiversity reduces food sources such as crops, fish, fruits, and livestock.

b) Health Problems

Many medicines come from plants and microorganisms. Biodiversity loss reduces future medical discoveries.

c) Loss of Livelihoods

Farmers, fishermen, forest dwellers, and tribal communities depend on biodiversity for income.

d) Poor Quality of Life

Reduced clean air, clean water, and natural beauty affect human well-being.

e) Cultural Loss

Many traditions, beliefs, and lifestyles are connected with plants and animals.

4. Discuss the importance of oceans in maintaining biodiversity. Explain the major problems faced by oceans and fisheries.

Answer (10 Marks)

Introduction

Oceans cover about **70% of the Earth's surface** and contain a rich variety of plants, animals, and microorganisms. They play a vital role in maintaining biodiversity, climate balance, and supporting human life. Oceans also provide important fishery resources for food and employment.

Importance of Oceans in Maintaining Biodiversity

1. Habitat for Marine Organisms

Oceans provide habitat for millions of species such as fish, whales, corals, plankton, seaweeds, and microorganisms.

2. Regulation of Climate

Oceans absorb heat and carbon dioxide, helping to regulate global temperature and climate.

3. Source of Food

Oceans provide fish, prawns, crabs, and other seafood that support global food security.

4. Oxygen Production

Marine plants and phytoplankton produce a significant portion of Earth's oxygen through photosynthesis.

5. Nutrient Cycling

Oceans help in recycling nutrients and maintaining ecological balance.

6. Economic Importance

They support fisheries, tourism, shipping, and coastal livelihoods.

Major Problems Faced by Oceans and Fisheries

1. Overfishing

Excessive fishing reduces fish populations faster than they can reproduce.

Effects:

- Decline in fish stocks
 - Threat to marine species
 - Loss of fishermen's income
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2. Bycatch

Unintentional capture of non-target species during fishing operations.

Examples:

- Dolphins
- Turtles
- Sharks

Effects:

- Death of useful and endangered species
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3. Marine Pollution

Pollution from plastics, oil spills, sewage, and industrial waste harms marine ecosystems.

Effects:

- Death of marine animals
 - Contaminated seafood
 - Destruction of coral reefs and habitats
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4. Climate Change

Rising temperatures and changing ocean conditions affect marine life.

Effects:

- Ocean warming
 - Coral bleaching
 - Ocean acidification
 - Migration of species
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5. Habitat Destruction

Destruction of mangroves, coral reefs, and coastal wetlands affects breeding grounds of fish.

5. What are Fisheries? Explain the types of fisheries and discuss the problems associated with overfishing, bycatch, marine pollution, and climate change.

Introduction

Fisheries refer to the activities related to the harvesting, breeding, processing, and management of fish and other aquatic organisms for food, employment, and commercial purposes. Fisheries are an important source of protein, livelihood, and economic development.

Types of Fisheries

1. Marine Fisheries

Marine fisheries involve catching fish and other organisms from seas and oceans.

Examples:

- Tuna
- Sardines
- Prawns
- Mackerel

Importance:

- Major source of seafood
 - Supports coastal communities
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2. Inland Fisheries

Inland fisheries are carried out in freshwater bodies such as rivers, lakes, ponds, reservoirs, and canals.

Examples:

- Carp
- Catfish
- Tilapia

Importance:

- Provides food and income in rural areas
 - Supports aquaculture activities
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3. Aquaculture (Fish Farming)

Cultivation of fish, prawns, and other aquatic species under controlled conditions.

Importance:

- Increases fish production
 - Reduces pressure on natural fisheries
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Problems Associated with Fisheries

1. Overfishing

Overfishing means catching fish at a rate faster than they can reproduce.

Effects:

- Decline in fish populations
 - Extinction risk of some species
 - Economic loss to fishermen
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2. Bycatch

Bycatch is the accidental capture of non-target species during fishing.

Examples:

- Dolphins
- Sea turtles
- Sharks

Effects:

- Death of endangered species
 - Disturbance of marine food chains
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3. Marine Pollution

Pollution of oceans by plastics, oil spills, sewage, and chemicals affects fisheries.

Effects:

- Poisoning of marine organisms
 - Reduced fish quality and quantity
 - Habitat destruction
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4. Climate Change

Climate change affects oceans and fish habitats.

Effects:

- Ocean warming changes fish migration patterns
- Coral bleaching destroys breeding grounds
- Ocean acidification affects shellfish
- Reduced fish productivity

6. Define Deforestation. Explain the major causes and environmental effects of deforestation.

Introduction

Forests are valuable natural resources that provide oxygen, regulate climate, conserve soil, and support biodiversity. Large-scale destruction of forests creates serious environmental problems.

Definition of Deforestation

Deforestation is the large-scale cutting, clearing, or removal of forests and trees for non-forest purposes such as agriculture, urbanization, industrialization, mining, and infrastructure development.

It results in permanent loss of forest cover and degradation of the environment.

Major Causes of Deforestation

1. Agricultural Expansion

Forests are cleared for cultivation, cattle grazing, and plantations.

Examples:

- Shifting cultivation
 - Commercial farming
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2. Logging and Timber Extraction

Trees are cut for timber, paper, furniture, and fuelwood.

Effects:

- Reduction in forest area
 - Loss of valuable tree species
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3. Urbanization and Infrastructure Development

Construction of roads, buildings, dams, industries, and railways leads to forest clearing.

4. Mining Activities

Extraction of minerals, coal, and other resources destroys forest land.

5. Forest Fires

Natural or human-caused fires burn large forest areas.

6. Population Growth

Increasing population creates demand for land, housing, fuel, and resources.

Environmental Effects of Deforestation

1. Loss of Biodiversity

Many plants and animals lose their natural habitat, leading to migration or extinction.

2. Soil Erosion

Removal of trees exposes soil to wind and rain, causing loss of fertile topsoil.

3. Disruption of Water Cycle

Forests help maintain rainfall and groundwater recharge. Their removal causes floods and droughts.

4. Increase in Carbon Dioxide

Trees absorb carbon dioxide. Cutting forests increases greenhouse gases and global warming.

5. Climate Change

Deforestation disturbs local and global climate patterns.

6. Desertification

Continuous forest loss may convert fertile land into dry barren land.

7. Loss of Natural Beauty and Resources

Forests provide medicinal plants, timber, and scenic beauty, which are reduced by deforestation.

7. Discuss the control measures of deforestation. Explain the role of afforestation, reforestation, sustainable forest management, and environmental laws.

Answer (10 Marks)

Introduction

Deforestation is the large-scale removal of forests for agriculture, urbanization, mining, and industrial purposes. It causes biodiversity loss, soil erosion, climate change, and ecological imbalance. Therefore, effective control measures are necessary to conserve forests and maintain environmental stability.

Control Measures of Deforestation

1. Afforestation

Afforestation means planting trees on barren or non-forest land to create new forests.

Role:

- Increases forest cover
- Prevents soil erosion
- Absorbs carbon dioxide
- Improves rainfall and climate conditions
- Provides habitat for wildlife

Example:

Tree plantation programs on unused lands.

2. Reforestation

Reforestation means replanting trees in areas where forests were cut or destroyed.

Role:

- Restores damaged forest ecosystems
- Conserves biodiversity
- Prevents floods and landslides
- Improves groundwater recharge

Example:

Replanting trees after forest fires or logging.

3. Sustainable Forest Management

It means using forest resources carefully without destroying their future productivity.

Practices:

- Selective cutting of trees
- Controlled harvesting
- Regeneration after cutting
- Prevention of illegal logging
- Community participation

Role:

- Ensures continuous supply of forest products
- Maintains ecological balance
- Protects wildlife habitats

4. Environmental Laws and Policies

Strong laws are necessary to regulate forest use and prevent destruction.

Role:

- Restricts illegal cutting of trees
- Protects reserved forests and wildlife areas
- Controls mining and industrial activities in forests
- Punishes violators

Examples:

- Forest Conservation Act
- Wildlife Protection Act
- Environmental Protection Act

5. Public Awareness and Education

- Educating people about the importance of forests
- Encouraging tree plantation drives
- Promoting eco-friendly lifestyles

6. Alternative Resources

- Use of LPG, solar energy, and biogas instead of fuelwood
- Recycling paper and wood products

8.Explain the International Dynamics of Biodiversity Conservation. Discuss the need for global cooperation in protecting biodiversity.

Introduction

Biodiversity is the variety of plants, animals, microorganisms, and ecosystems on Earth. It is essential for ecological balance, food security, medicine, and sustainable development. Since environmental problems cross national boundaries, biodiversity conservation requires **international cooperation** and joint global efforts.

International Dynamics of Biodiversity Conservation

International dynamics refers to the collective actions, treaties, organizations, and policies adopted by countries to conserve biodiversity worldwide.

1. Convention on Biological Diversity (CBD)

The **Convention on Biological Diversity (CBD)** was adopted in 1992 at the Rio Earth Summit.

Main Objectives:

- Conservation of biodiversity
 - Sustainable use of biological resources
 - Fair and equitable sharing of benefits from genetic resources
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2. International Union for Conservation of Nature (IUCN)

The **IUCN** is a global organization working for nature conservation.

Functions:

- Publishes the Red List of threatened species
 - Supports wildlife conservation programs
 - Advises governments on environmental policies
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3. Protected Areas and Wildlife Conservation

Many countries establish protected areas such as:

- **National Parks**
- **Wildlife Sanctuaries**
- **Biosphere Reserves**

These areas help conserve habitats and endangered species.

4. International Agreements

Other important global agreements include:

- **CITES** – Controls illegal trade in wildlife species
 - **Ramsar Convention** – Protects wetlands
 - **UNESCO Programmes** – Conservation of biosphere reserves
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Need for Global Cooperation in Protecting Biodiversity

1. Transboundary Nature of Ecosystems

Many forests, rivers, oceans, and migratory species extend across countries.

2. Climate Change

Global warming affects biodiversity worldwide and needs joint action.

3. Control of Illegal Wildlife Trade

International cooperation is needed to stop poaching and smuggling.

4. Sharing Scientific Knowledge

Countries can share research, technology, and conservation methods.

5. Financial and Technical Support

Developing countries may need assistance for biodiversity conservation.

6. Sustainable Development

Global cooperation helps balance economic growth with environmental protection.

9. Write a note on Convention on Biological Diversity (CBD), IUCN, and protected areas such as national parks, wildlife sanctuaries, and biosphere reserves.

Introduction

Biodiversity is essential for ecological balance and sustainable development. To protect biodiversity, several international organizations, agreements, and protected areas have been established throughout the world.

1. Convention on Biological Diversity (CBD)

The **Convention on Biological Diversity (CBD)** is an international agreement adopted at the **Rio Earth Summit in 1992**.

Main Objectives of CBD

- 1. Conservation of biodiversity**
Protection of ecosystems, species, and genetic resources.
- 2. Sustainable use of biological resources**
Proper use of resources without depletion.
- 3. Fair sharing of benefits**
Equal sharing of benefits arising from genetic resources and biotechnology.

Importance of CBD

- Encourages countries to prepare biodiversity action plans
- Promotes conservation programs
- Supports sustainable development

2. International Union for Conservation of Nature (IUCN)

The **IUCN** is a global organization working for conservation of nature and natural resources.

Functions of IUCN

- Publishes the **Red List of Threatened Species**
- Provides scientific data on endangered species
- Supports wildlife and habitat conservation
- Advises governments on environmental policies

Importance

- Helps identify species at risk of extinction
- Promotes global conservation awareness

3. Protected Areas

Protected areas are regions reserved for conserving wildlife, plants, and ecosystems.

a) National Parks

These are protected areas established to conserve wildlife and natural habitats.

Features:

- No hunting or commercial activities allowed

- Strict protection by government

Examples:

- Bandipur National Park
 - Kaziranga National Park
-

b) Wildlife Sanctuaries

Areas established mainly for the protection of animals and birds.

Features:

- Limited human activities may be allowed
- Protection of endangered species

Examples:

- Ranganathittu Bird Sanctuary
 - Vedanthangal Bird Sanctuary
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c) Biosphere Reserves

Large protected areas aimed at conserving biodiversity along with sustainable use by local communities.

Features:

- Core zone, buffer zone, transition zone
- Combines conservation and development

Examples:

- Nilgiri Biosphere Reserve
- Nanda Devi Biosphere Reserve

10. “Biodiversity conservation is essential for sustainable development.” Justify the statement with suitable examples.

Introduction

Sustainable development means meeting the needs of the present generation without compromising the ability of future generations to meet their own needs. **Biodiversity conservation** is essential for sustainable development because biodiversity provides food, water, medicine, raw materials, ecological balance, and economic benefits.

Why Biodiversity Conservation is Essential for Sustainable Development

1. Maintains Ecological Balance

Biodiversity helps maintain food chains, nutrient cycles, and ecosystem stability.

Example:

Forests support birds, insects, and animals, maintaining balance in nature.

2. Provides Food Security

Many crops, fruits, fish, and livestock depend on biodiversity.

Example:

Different varieties of rice and wheat ensure food supply and resistance to diseases.

3. Source of Medicines

Many medicines are obtained from plants, animals, and microorganisms.

Example:

Neem, Tulsi, and Cinchona are used in medicine production.

4. Supports Agriculture

Pollinators like bees and butterflies help in crop production.

Example:

Pollination increases yield of fruits and vegetables.

5. Climate Regulation

Forests, wetlands, and oceans absorb carbon dioxide and regulate climate.

Example:

Amazon rainforest acts as a carbon sink reducing global warming.

6. Economic Benefits

Biodiversity supports industries such as fisheries, forestry, tourism, and agriculture.

Example:

Wildlife tourism in national parks generates income and employment.

7. Disaster Protection

Natural ecosystems reduce impacts of floods, storms, and soil erosion.

Example:

Mangrove forests protect coastal areas from cyclones and tsunamis.

8. Preserves Resources for Future Generations

Conservation ensures continuous availability of natural resources.

Example:

Sustainable fishing helps maintain fish stocks for future use.

Suitable Examples of Conservation

- National Parks and Wildlife Sanctuaries protect species.
- Afforestation restores forests.
- Organic farming conserves soil biodiversity.
- Biosphere reserves combine development and conservation.